Class XII Session 2025-26
Subject - Chemistry
Sample Question Paper - 3

Time Allowed: 3 hours

General Instructions:

1. This Question paper contains - five sections A, Each section is compulsory. However, there are internal choices in some questions.
2. Section A has 20 MCQ's and 02 Assertion-Reason based questions of 1 mark each.

1. **Section A**

2. If $A = \begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is symmetric then what is x equal to? **[1]**

   a) 5 b) 2

   c) 3 d) -1

3. If A and B are square matrices of order 3, such that Det.A = –1, Det.B =3 then, the determinant of 3AB is equal to **[1]**

   a) –9 b) –27

   c) -81 d) 81

4. The inverse of the matrix $\begin{vmatrix} 2 & -1 \\ 3 & 4 \end{vmatrix}$ is **[1]**

   a) $\begin{vmatrix} 4/11 & 1/11 \\ -3/11 & 2/11 \end{vmatrix}$ b) $\begin{vmatrix} 4 & 1 \\ -3 & 2 \end{vmatrix}$

   c) $\begin{vmatrix} 2 & 1 \\ 0 & 3 \end{vmatrix}$ d) $\begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$

5. If $x = a(\cos\theta + \theta\sin\theta)$ and $y = a(\sin\theta - \theta\cos\theta)$ $\frac{dy}{dx} = ?$ **[1]**

   a) tan θ b) a cot θ

   c) a tan θ d) cot θ

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6. The general solution of the differential equation $e^x\,dy + (y\,e^x + 2x)\,dx = 0$ is **[1]**

a) $x\,e^y + y^2 = C$

b) $y\,e^y + x^2 = C$

c) $x\,e^y + x^2 = C$

d) $y\,e^x + x^2 = C$

7. Corner points of the feasible region for an LPP are (0, 2), (3, 0), (6, 0), (6, 8) and (0, 5).Let F = 4x + 6y be the objective function. The Minimum value of F occurs at **[1]**

a) any point on the line segment joining the points (0, 2) and (3, 0).

b) (0, 2) only

c) the mid-point of the line segment joining the points (0, 2) and (3, 0) only

d) (3, 0) only

8. The value of $\sin^{-1}\left(\cos\frac{3\pi}{5}\right)$ is **[1]**

a) $\frac{-3\pi}{5}$

b) $\frac{-\pi}{10}$

c) $\frac{\pi}{10}$

d) $\frac{3\pi}{5}$

9. $\int \frac{dx}{\sqrt{x-x^2}} = ?$ **[1]**

a) $\sin^{-1}(2x + 1) + C$

b) $\sin^{-1}(2x - 1) + C$

c) $\sin^{-1}(x + 1) + C$

d) $\sin^{-1}(x - 1) + C$

10. If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then $A^4 =$ **[1]**

a) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

d) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$

11. Maximize $Z = 100x + 120y$, subject to constraints $2x + 3y \le 30$, $3x + y \le 17$, $x \ge 0$, $y \ge 0$. **[1]**

a) 1200

b) 1260

c) 1280

d) 1300

12. What is the vector perpendicular to both the vectors $\boldsymbol{i} - \boldsymbol{j}$ and $\boldsymbol{i}$? **[1]**

a) $\boldsymbol{j}$

b) $\boldsymbol{k}$

c) $-\boldsymbol{j}$

d) $\boldsymbol{i}$

13. The value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ = 3A - B then the values of A and B are: **[1]**

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a) A = 2abc, B = a + b + c

b) A = 3abc, B = a + b + c **[1]**

c) A = 0, B = $a^2 + b^2 + c^2$

d) A = abc, B = $a^3 + b^3 + c^3$

14. If $P(A \cap B) = \frac{1}{8}$ and $P(A) = \frac{3}{4}$, then $P\left(\frac{B}{A}\right)$ is equal to: **[1]** **[1]**

a) $\frac{1}{2}$

b) $\frac{2}{3}$

c) $\frac{1}{3}$

d) $\frac{1}{6}$

15. The degree **15.** of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + \sin\left(\frac{dy}{dx}\right) + 1 = 0$ is

a) Not defined

b) 1

c) 2

d) 3

16. The position vectors of three consecutive vertices of a parallelogram ABCD are $A(4i + 2j - 6k)$, B $(5i - 3j + k)$ and $C(12i + 4j + 5k)$. The position vector of D is given by

a) $-11i - 9j + 2k$

b) $-3i - 5j - 10k$

c) $11i + 9j - 2k$

d) $21i + 3j$

17. The derivative of sin x w.r.t. cos x is **[1]**

a) cosec x

b) tan x

c) -cot x

d) cot x

18. The lines l_1 and l_2 intersect. The shortest distance between them is **[1]**

a) zero

b) positive

c) infinity

d) negative

19. **Assertion (A):** A particle moving in a straight line covers a distance of x cm in t second, where $x = t^3 + 3t^2 - 6t + 18$. The velocity of particle at the end of 3 seconds is 39 cm/s. **[1]**

Reason (R): Velocity of the particle at the end of 3 seconds is $\frac{dx}{dt}$ at t = 3.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

20. **Assertion (A):** The number of onto functions from a set P containing 5 elements to a set Q containing 2 elements is 30. **[1]**

Reason (R): Number of onto functions from a set containing m elements to a set containing n elements is n^m.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

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